GRIDMASTER

Power Grid Emergency Simulator

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The Problem

GridMaster is an interactive power grid simulator that lets engineering students and trainees practice responding to real electrical grid emergencies without risk to any real infrastructure.

Key Features

1. Live grid visualization: a real-time diagram showing power flow, line loading percentages, and generator output across all 14 buses, with color-coded warnings as conditions change.

2. Five emergency scenarios: generator trip, load spike, line outage, generation crisis, and N-1 cascade failure, each requiring a different operator response.

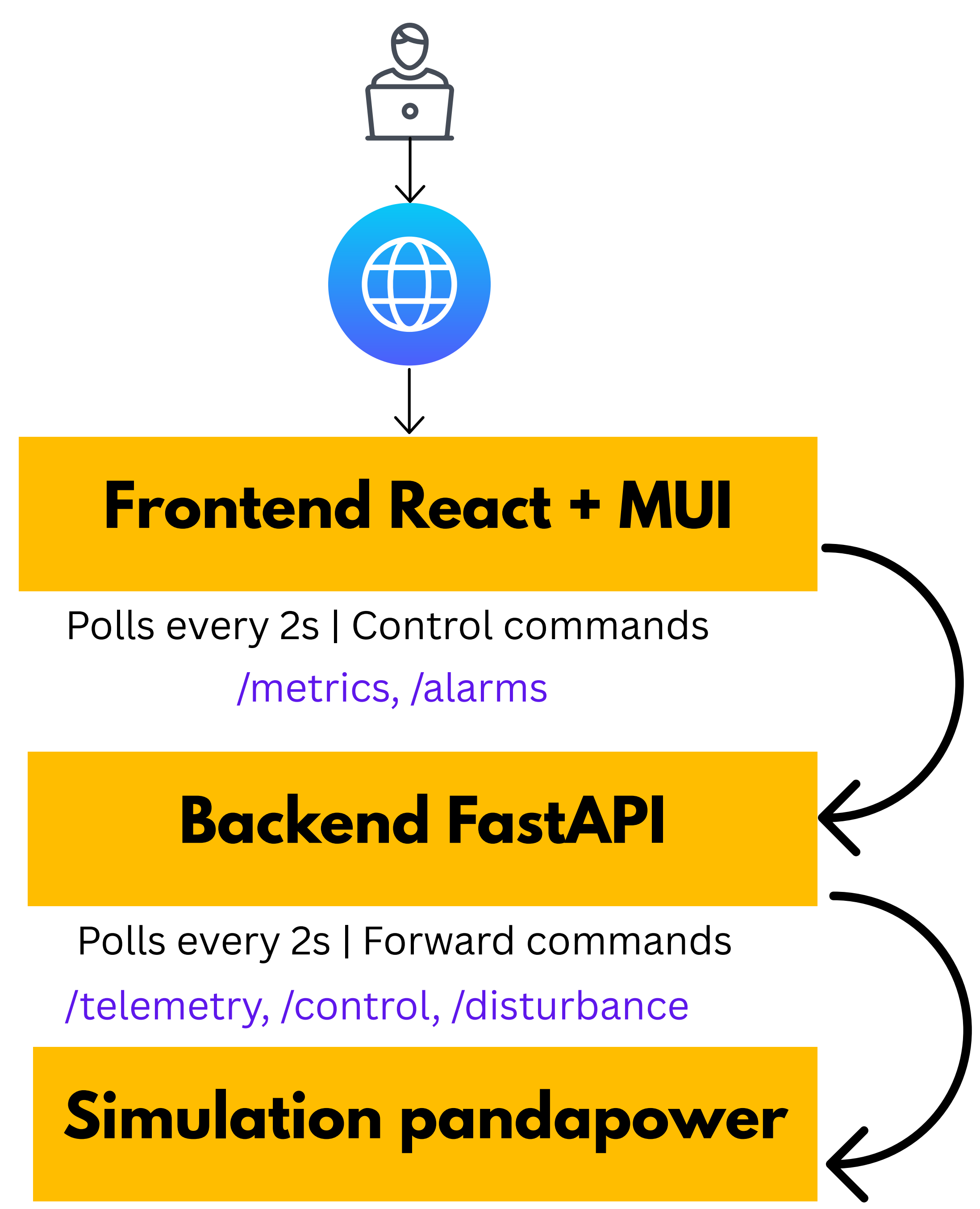
3. Active alarm system: severity-ranked alarms fire automatically when thresholds are crossed, with operator acknowledgment and built-in guidance on how to respond.

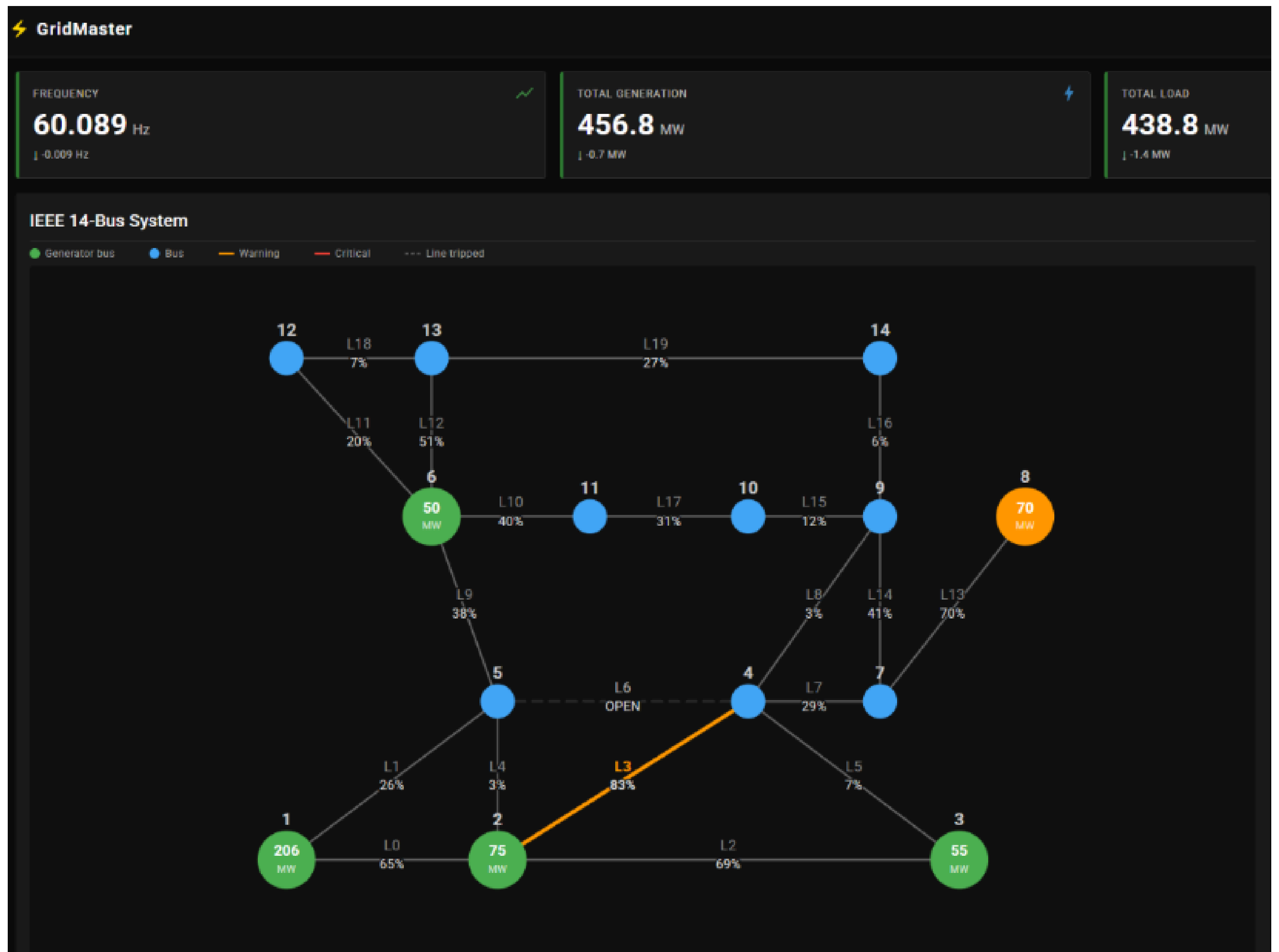
4. Manual operator controls: adjust generator output in real time, trip or restore individual transmission lines, and watch the grid respond.

Why This Exists

Power grid operators make high-stakes decisions under pressure: a single misstep can cascade into widespread outages. But you can't practice on a live grid. Traditional training relies on textbooks and static diagrams that don't capture how a real system behaves when things go wrong. GridMaster fills that gap: it runs actual power flow calculations on an industry-standard IEEE 14-bus network and lets users trigger, observe, and resolve real failure scenarios in real time.

Architecture Diagram





Tech Stack